

Clinical Effectiveness

METIMMOX-1 Economic Evaluation

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1 Overview

This report presents clinical effectiveness results from the METIMMOX-1 trial, evaluating biomarker-guided treatment strategies for selecting immunotherapy candidates in metastatic MSS/pMMR colorectal cancer.

The analysis uses **Firth-corrected Cox proportional hazards regression** with profile likelihood confidence intervals to assess treatment effect heterogeneity across two pre-treatment biomarkers identified by the causal DAG analysis:

- **CRP** (C-reactive protein): Low CRP (<5 mg/L) at baseline — a pre-treatment prognostic and potentially predictive marker
- **TMB/BRAF**: High tumor mutational burden (9 mut/MB) or BRAF mutation — a pre-treatment genomic marker

Model specification is motivated by the causal DAG (see `dag.qmd` and `dag_associations.qmd`).

The DAG encodes treatment (T) as randomized with no parents, so no confounding adjustment is needed for causal identification; Age and Sex are included as precision covariates. TLR (Tumor Lesion Reduction) is encoded as a post-randomization intermediate variable ($T \rightarrow \text{TLR} \rightarrow \text{PFS}$): it is not a pre-treatment baseline characteristic and conditioning on it as a covariate would block part of the treatment effect pathway. TLR is therefore excluded from the primary survival models and examined separately as a descriptive endpoint.

The **unified DAG-informed model** is:

$$\text{Surv}(\text{time}, \text{event}) \sim \text{Age} + \text{sex} + \text{Rx} + \text{CRP} + \text{TMB/BRAF} + \text{CRP} \times \text{Rx} + \text{TMB/BRAF} \times \text{Rx}$$

Ridge regression (L2-penalized Cox) is applied as a sensitivity analysis to assess the stability of the interaction estimates under shrinkage. TLR is reported in a separate exploratory section.

2 Methodological Notes

2.1 Software Packages

This analysis uses the following R packages:

- `coxphf` version 1.13.4: Firth-corrected Cox regression with profile likelihood CIs

- `survival` version 3.7.0: Cox regression and Schoenfeld residual diagnostics
- `glmnet` version 4.1.8: Ridge regression sensitivity analysis
- `vcd` version 1.4.13: Cramér’s V for biomarker correlation assessment

2.2 Firth-Corrected Cox Regression

We use Firth’s penalized maximum likelihood method for Cox regression, which addresses:

1. **Monotone likelihood:** When a covariate perfectly separates events from non-events
2. **Small sample bias:** Reduces bias in coefficient estimates with sparse data
3. **Unstable confidence intervals:** Profile likelihood CIs are more reliable than Wald-based CIs in small samples

Statistical inference is based on **penalized likelihood ratio tests (PLRT)**, which are more appropriate than Wald tests for testing treatment-biomarker interactions in this setting.

2.3 Ridge Regression Sensitivity Analysis

We use Ridge regression as a sensitivity analysis to assess the stability of our interaction estimates. Ridge regression (also called L2-penalized regression or Tikhonov regularization) belongs to the family of regularized or penalized regression methods. In survival analysis, this is implemented as a penalized Cox model.

How Ridge regression works:

- Ridge adds a penalty term that discourages large coefficients. The bigger a coefficient tries to get, the more the model resists, pulling estimates toward zero (HR toward 1.0).
- Unlike LASSO ($\alpha = 1$), which sets some coefficients exactly to zero and performs variable selection, Ridge ($\alpha = 0$) shrinks all coefficients but retains all predictors.
- The degree of shrinkage is controlled by the tuning parameter λ , which is selected via **cross-validation (CV)**: the data are repeatedly split into training and validation sets, and the λ that produces the best predictive performance is chosen.

Why not LASSO or Elastic Net?

We determined that LASSO and elastic net are **not appropriate** for this analysis:

1. **Correlated biomarkers:** LASSO arbitrarily selects one of correlated predictors while setting others to zero, which does not reflect the clinical question of comparing ALL biomarker strategies.
2. **Hierarchy principle violation:** LASSO may select interaction terms (e.g., `crp:Rx`) while setting the main effect (`crp`) to zero, producing models that are difficult to interpret clinically.
3. **Clinical objective:** Our goal is to estimate treatment effect heterogeneity across ALL biomarkers simultaneously, not to select a “winning” biomarker.

Elastic net ($\alpha = 0.1$ – 0.3) represents a compromise but does not fully address these concerns.

Interpreting Ridge results:

Ridge regression is not replacing the Firth estimates—it provides a reality check. When Ridge substantially shrinks an interaction coefficient toward zero while Firth estimates a large effect, this indicates the effect is too unstable to trust for practical use. The CV-selected λ reflects how much regularization is needed for reasonable out-of-sample prediction given the sample size and collinearity structure.

3 Data Preparation

3.1 Sample Characteristics

Table 1: Sample Characteristics by Biomarker Status

Characteristic	Overall	CRP		TLR		TMB/BRAF	
		Positive	Negative	Positive	Negative	Positive	Negative
Patients, n	65	22	43	41	24	29	36
Demographics							
Age, mean (SD)	64.0 (10.0)	64.3 (10.3)	63.8 (10.0)	65.0 (9.9)	62.2 (10.2)	64.6 (10.4)	63.5 (9.8)
Male	30 (46.2%)	12 (54.5%)	18 (41.9%)	19 (46.3%)	11 (45.8%)	14 (48.3%)	16 (44.4%)
Female	35 (53.8%)	10 (45.5%)	25 (58.1%)	22 (53.7%)	13 (54.2%)	15 (51.7%)	20 (55.6%)
Treatment arm							
Control (FLOX alone)	29 (44.6%)	5 (22.7%)	24 (55.8%)	22 (53.7%)	7 (29.2%)	13 (44.8%)	16 (44.4%)
Experimental (FLOX/nivolumab)	36 (55.4%)	17 (77.3%)	19 (44.2%)	19 (46.3%)	17 (70.8%)	16 (55.2%)	20 (55.6%)
Outcome events							
Deaths (OS events)	56 (86.2%)	16 (72.7%)	40 (93.0%)	32 (78.0%)	24 (100.0%)	22 (75.9%)	34 (94.4%)
Progressions (PFS events)	48 (73.8%)	14 (63.6%)	34 (79.1%)	27 (65.9%)	21 (87.5%)	19 (65.5%)	29 (80.6%)
Biomarker prevalence							
CRP-positive (low CRP)	22 (33.8%)	22 (100.0%)	0 (0.0%)	18 (43.9%)	4 (16.7%)	11 (37.9%)	11 (30.6%)
TLR-positive (early response)	41 (63.1%)	18 (81.8%)	23 (53.5%)	41 (100.0%)	0 (0.0%)	18 (62.1%)	23 (63.9%)
TMB/BRAF-positive	29 (44.6%)	11 (50.0%)	18 (41.9%)	18 (43.9%)	11 (45.8%)	29 (100.0%)	0 (0.0%)

Note: Percentages within each column use the column subgroup as denominator. CRP/TLR/TMB-BRAF rows in the corresponding subgroup column equal 100% by definition.

3.2 Biomarker Correlations

Table 2: Biomarker Correlations (Cramér’s V)

Biomarker Pair	Cramér’s V	Interpretation
CRP vs TLR	0.278	Weak
CRP vs TMB/BRAF	0.077	Negligible
TLR vs TMB/BRAF	0.019	Negligible

Note: Cramér’s V interpretation: <0.1 = Negligible, $0.1\text{--}0.3$ = Weak, >0.3 = Moderate/Strong. The observed correlations are weak to negligible. While modest, any correlation among candidate biomarkers supports our decision not to use LASSO, which would arbitrarily select among correlated predictors.

4 Proportional Hazards Assumption

4.1 Schoenfeld Residual Tests

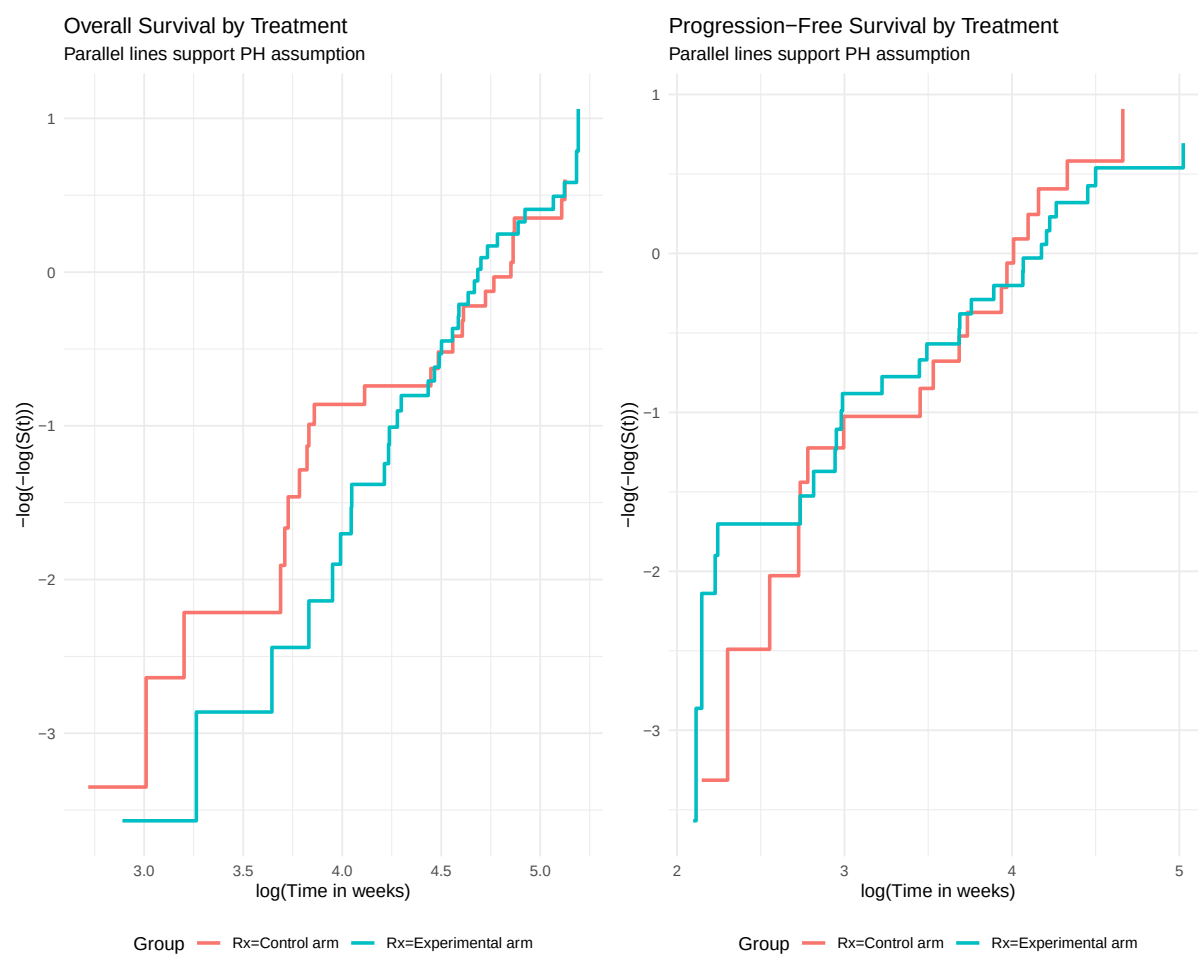
The proportional hazards assumption was tested using scaled Schoenfeld residuals. A significant p-value indicates potential violation of the PH assumption for that covariate.

Table 3: Schoenfeld Residual Test for Proportional Hazards Assumption

Covariate	χ^2 (OS)	p-value	χ^2 (PFS)	p-value
Age	1.85	0.174	0.50	0.480
sex	0.10	0.751	0.04	0.847
Rx	3.15	0.076	0.04	0.841
crp_num	0.89	0.345	3.22	0.073
tmb_braf_num	1.01	0.315	5.33	0.021
Rx:crp_num	2.26	0.133	1.21	0.272
Rx:tmb_braf_num	0.57	0.450	4.01	0.045
GLOBAL	7.32	0.396	19.60	0.007

Note: GLOBAL is the omnibus test for all covariates. Individual covariate tests help identify specific PH violations.

4.2 Log-Log Survival Plots



Interpretation: Approximately parallel lines on the log-log plot support the proportional hazards assumption. Crossing or converging lines suggest time-varying effects.

5 Primary Analysis: Firth-Corrected Cox Models

5.1 Standard Cox vs Firth

The standard Cox model is the unpenalized maximum-likelihood reference. Divergence between standard Cox and Firth estimates flags small-sample or near-separation instability that Firth corrects.

Table 4: Unified Model - Overall Survival: Standard Cox vs Firth

Term	Standard Cox HR (95% CI)	Firth HR (95% CI)
Treatment x Biomarker Interactions		
CRP x Rx	0.88 (0.20-3.85)	0.78 (0.21-3.61)
TMB/BRAF x Rx	0.95 (0.30-2.97)	0.93 (0.30-2.85)
Main Effects and Covariates		
Rx (Experimental)	1.69 (0.79-3.63)	1.70 (0.81-3.67)
CRP	0.45 (0.13-1.55)	0.52 (0.13-1.47)
TMB/BRAF	0.91 (0.40-2.10)	0.94 (0.41-2.12)
Age	1.00 (0.97-1.03)	1.00 (0.97-1.03)
Sex	1.59 (0.90-2.81)	1.58 (0.90-2.79)

Table 5: Unified Model - Progression-Free Survival: Standard Cox vs Firth

Term	Standard Cox HR (95% CI)	Firth HR (95% CI)
Treatment x Biomarker Interactions		
CRP x Rx	0.38 (0.06-2.34)	0.33 (0.07-2.15)
TMB/BRAF x Rx	0.61 (0.16-2.38)	0.60 (0.16-2.28)
Main Effects and Covariates		
Rx (Experimental)	2.04 (0.86-4.81)	2.01 (0.89-4.84)
CRP	0.84 (0.18-3.95)	1.00 (0.19-3.60)
TMB/BRAF	0.74 (0.27-1.99)	0.77 (0.28-2.01)
Age	0.99 (0.95-1.02)	0.99 (0.96-1.02)
Sex	1.21 (0.65-2.26)	1.20 (0.65-2.26)

5.2 Unified Model Results

The unified DAG-informed model includes CRP and TMB/BRAF as pre-treatment biomarkers with their treatment interactions, adjusted for Age and Sex.

Formula: $\text{Surv}(\text{time}, \text{event}) \sim \text{Age} + \text{sex} + \text{Rx} + \text{CRP} + \text{TMB/BRAF} + \text{CRP:Rx} + \text{TMB/BRAF:Rx}$

5.2.1 Overall Survival

Table 6: Unified Model: Overall Survival - Full Coefficient Table

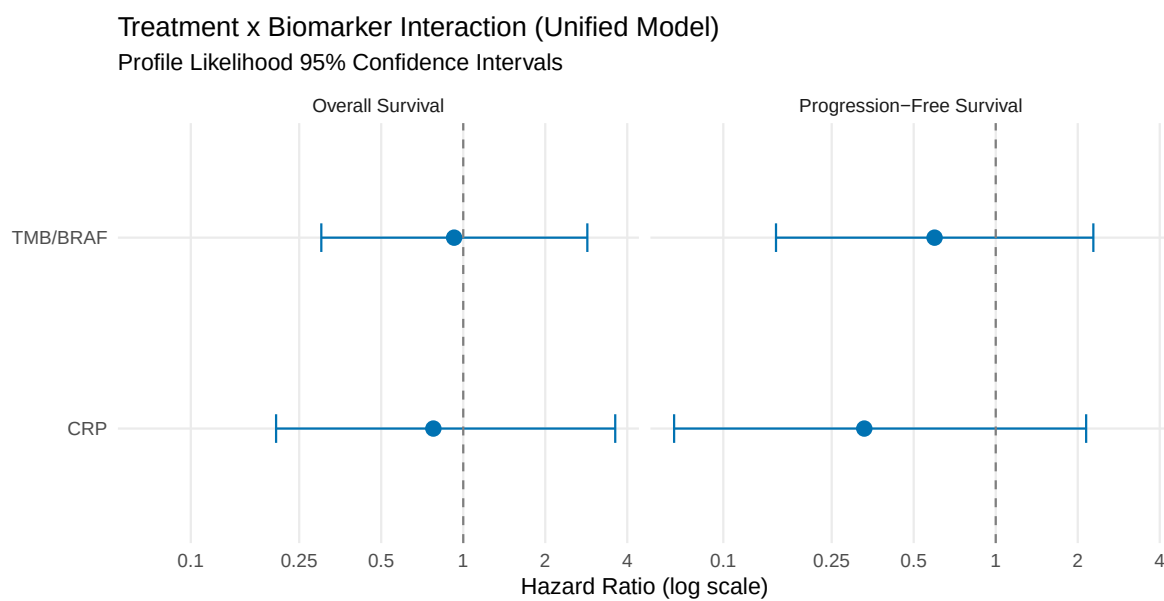
Term	HR (95% CI)	PLRT p-value
Treatment x Biomarker Interactions		
RxExperimental arm:crp_num	0.78 (0.21-3.61)	0.728
RxExperimental arm:tmb_braf_num	0.93 (0.30-2.85)	0.893
Main Effects and Covariates		
Age	1.00 (0.97-1.03)	0.987
sex1	1.58 (0.90-2.79)	0.110
RxExperimental arm	1.70 (0.81-3.67)	0.164
crp_num	0.52 (0.13-1.47)	0.233
tmb_braf_num	0.94 (0.41-2.12)	0.891

5.2.2 Progression-Free Survival

Table 7: Unified Model: Progression-Free Survival - Full Coefficient Table

Term	HR (95% CI)	PLRT p-value
Treatment x Biomarker Interactions		
RxExperimental arm:crp_num	0.33 (0.07-2.15)	0.224
RxExperimental arm:tmb_braf_num	0.60 (0.16-2.28)	0.447
Main Effects and Covariates		
Age	0.99 (0.96-1.02)	0.386
sex1	1.20 (0.65-2.26)	0.553
RxExperimental arm	2.01 (0.89-4.84)	0.096
crp_num	1.00 (0.19-3.60)	0.998
tmb_braf_num	0.77 (0.28-2.01)	0.594

5.3 Forest Plot



Interpretation: $HR < 1$ indicates that biomarker-positive patients derive greater benefit from experimental treatment (reduced hazard of death/progression). The dashed line at $HR = 1$ represents no differential treatment effect.

6 Sensitivity Analysis: Ridge Regression

Ridge regression (L2-penalized Cox, $\alpha = 0$) shrinks all coefficients toward zero but does not eliminate any, providing a sensitivity check for the stability of the interaction estimates. The same unified model formula is used; only the main effects (CRP and TMB/BRAF) are penalized.

Table 8: Overall Survival: Firth vs Ridge Regression

Interaction	Firth (95% CI)	Ridge (point est.)
CRP x Rx	0.78 (0.21-3.61)	0.39 (SE: 2.14)
TMB/BRAF x Rx	0.93 (0.30-2.85)	0.87 (SE: 2.04)

Note: Ridge: L2 shrinkage on main effects. Substantial shrinkage vs Firth = estimate sensitive to regularization.

Table 9: Progression-Free Survival: Firth vs Ridge Regression

Interaction	Firth (95% CI)	Ridge (point est.)
CRP x Rx	0.33 (0.07-2.15)	0.32 (SE: 2.15)
TMB/BRAF x Rx	0.60 (0.16-2.28)	0.45 (SE: 2.05)

Note: Ridge applies L2 shrinkage to main effects only.

Interpretation: Firth provides the best unbiased point estimate given the sample size and data structure. Ridge applies L2 shrinkage toward zero (HR toward 1.0). Comparing the two methods reveals estimate stability: if Firth and Ridge agree closely, the estimate is robust; if Ridge substantially shrinks the interaction toward null, the effect is sensitive to regularization in this sample.

7 Exploratory: TLR as a Predictive Biomarker (Responder Analysis)

This section presents an **exploratory responder analysis** that includes TLR (Tumor Lesion Reduction) as an effect modifier in a Cox model with the same structural form as the primary analysis. Because TLR is measured post-randomization (at the first on-treatment CT scan), this is not a pre-treatment patient-selection analysis: TLR status is itself influenced by treatment, and the resulting TLR x Rx interaction is informative only as a hypothesis-generating, responder-stratified contrast. It cannot be used to guide treatment decisions at baseline. The primary causal-DAG-informed analysis remains the inferential anchor for predictive biomarker claims.

The model adjusts for the pre-treatment biomarkers **CRP** and **TMB/BRAF** as prognostic main effects (their treatment interactions belong to the primary analysis and are not re-estimated here). In the ridge sensitivity model, CRP and TMB/BRAF remain **unpenalized** (well-established prognostic adjustment covariates), while the **TLR main effect and TLR x Rx interaction are penalized**. These are the post-randomization terms whose stability we want to stress-test against the small effective sample size and the partially tautological TLR-PFS association.

Formula: $\text{Surv}(\text{time}, \text{event}) \sim \text{Age} + \text{sex} + \text{Rx} + \text{CRP} + \text{TMB/BRAF} + \text{TLR} + \text{TLR:Rx}$

7.1 Descriptive Context

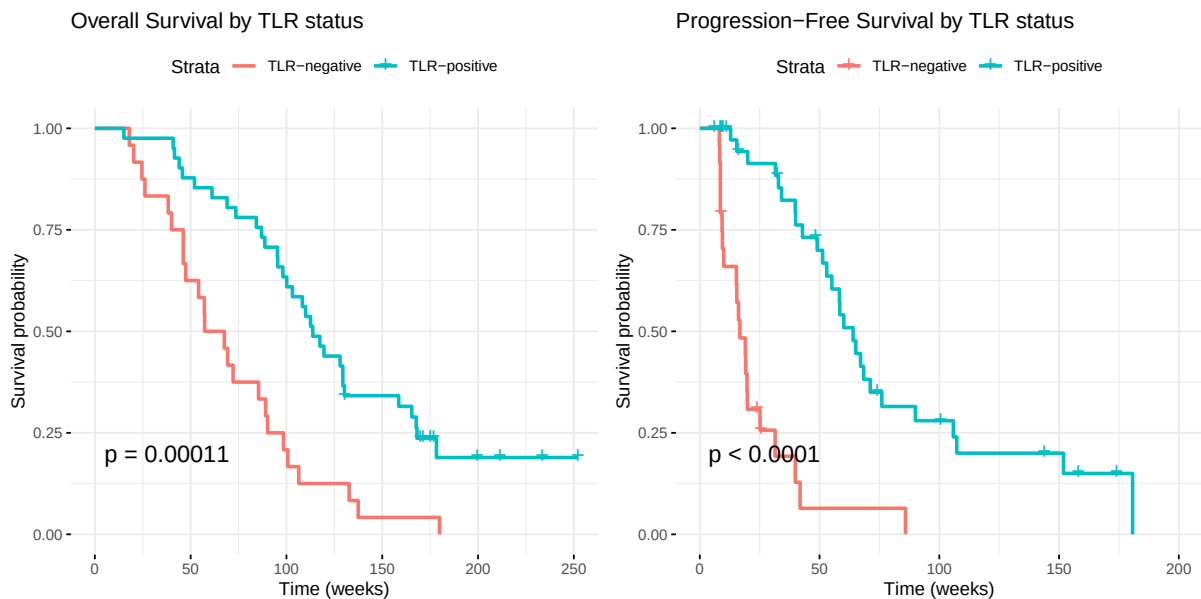
7.1.1 TLR Prevalence by Treatment Arm

Table 10: TLR Prevalence by Treatment Arm (Descriptive)

Treatment Arm	N	TLR-positive	%
Control (FLOX)	29	22	75.9
Experimental (FLOX/nivo)	36	19	52.8

Note: TLR = Tumour Lesion Reduction $\geq 10\%$ at first CT scan (post-treatment).

7.1.2 Kaplan-Meier Curves Stratified by TLR



Note: These curves are purely descriptive and stratify by a post-treatment response variable. The observed difference in survival by TLR status reflects treatment response, not a pre-treatment patient characteristic.

7.2 Standard Cox vs Firth

The standard Cox model is the unpenalized maximum-likelihood reference. Divergence between standard Cox and Firth estimates flags small-sample or near-separation instability that Firth corrects.

Table 11: TLR Responder Model - Overall Survival: Standard Cox vs Firth

Term	Standard Cox HR (95% CI)	Firth HR (95% CI)
TLR Terms		
TLR x Rx	2.43 (0.74-7.95)	2.47 (0.75-7.75)
TLR (main effect)	0.21 (0.08-0.54)	0.21 (0.08-0.55)
Adjustment Covariates		
Rx (Experimental)	0.60 (0.23-1.55)	0.59 (0.24-1.55)
CRP	0.52 (0.25-1.06)	0.53 (0.25-1.05)
TMB/BRAF	0.86 (0.48-1.52)	0.86 (0.48-1.52)
Age	1.01 (0.98-1.04)	1.01 (0.98-1.04)
Sex	1.48 (0.81-2.68)	1.47 (0.82-2.67)

Table 12: TLR Responder Model - Progression-Free Survival: Standard Cox vs Firth

Term	Standard Cox HR (95% CI)	Firth HR (95% CI)
TLR Terms		
TLR x Rx	0.73 (0.19-2.76)	0.75 (0.20-2.66)
TLR (main effect)	0.17 (0.06-0.50)	0.17 (0.06-0.52)
Adjustment Covariates		
Rx (Experimental)	1.41 (0.46-4.27)	1.35 (0.48-4.22)
CRP	0.41 (0.18-0.96)	0.43 (0.18-0.96)
TMB/BRAF	0.48 (0.24-0.99)	0.50 (0.24-0.99)
Age	1.00 (0.96-1.03)	1.00 (0.96-1.03)
Sex	1.23 (0.61-2.47)	1.22 (0.62-2.45)

7.3 Firth-Corrected Cox Model

7.3.1 Overall Survival

Table 13: TLR Responder Model: Overall Survival - Full Coefficient Table

Term	HR (95% CI)	PLRT p-value
Treatment x Biomarker Interaction		
RxExperimental arm:tlr_num	2.47 (0.75-7.75)	0.135
Main Effects and Covariates		
Age	1.01 (0.98-1.04)	0.626
sex1	1.47 (0.82-2.67)	0.201
RxExperimental arm	0.59 (0.24-1.55)	0.270
crp_num	0.53 (0.25-1.05)	0.071
tmb_braf_num	0.86 (0.48-1.52)	0.609
tlr_num	0.21 (0.08-0.55)	0.002

7.3.2 Progression-Free Survival

Table 14: TLR Responder Model: Progression-Free Survival - Full Coefficient Table

Term	HR (95% CI)	PLRT p-value
Treatment x Biomarker Interaction		
RxExperimental arm:tlr_num	0.75 (0.20-2.66)	0.660
Main Effects and Covariates		
Age	1.00 (0.96-1.03)	0.787
sex1	1.22 (0.62-2.45)	0.570
RxExperimental arm	1.35 (0.48-4.22)	0.576
crp_num	0.43 (0.18-0.96)	0.039
tmb_braf_num	0.50 (0.24-0.99)	0.046
tlr_num	0.17 (0.06-0.52)	0.003

7.4 Ridge Sensitivity Analysis (TLR Terms Penalized)

Ridge regression is applied here with a targeted penalty: only the TLR main effect and the TLR x Rx interaction are L2-shrunk. CRP, TMB/BRAF, Age, sex, and Rx are estimated without penalty, since they enter the model as adjustment covariates whose effects are not the inferential target.

Table 15: TLR Responder Model - Overall Survival: Firth vs Ridge, all coefficients

Term	Firth (95% CI)	Ridge HR (SE)
TLR Terms (Ridge-penalized)		
TLR x Rx	2.47 (0.75-7.75)	2.12 (SE: 0.58)
TLR (main effect)	0.21 (0.08-0.55)	0.24 (SE: 0.47)
Adjustment Covariates (Ridge-unpenalized)		
Age	1.01 (0.98-1.04)	1.01 (SE: 0.02)
Sex	1.47 (0.82-2.67)	1.49 (SE: 0.30)
Rx (Experimental)	2.47 (0.75-7.75)	0.66 (SE: 0.48)
CRP	0.53 (0.25-1.05)	0.52 (SE: 0.36)
TMB/BRAF	0.86 (0.48-1.52)	0.86 (SE: 0.29)

Note: Ridge applies L2 shrinkage to the TLR main effect and the TLR x Rx interaction only; Age, sex, Rx, CRP, and TMB/B

Table 16: TLR Responder Model - Progression-Free Survival: Firth vs Ridge, all coefficients

Term	Firth (95% CI)	Ridge HR (SE)
TLR Terms (Ridge-penalized)		
TLR x Rx	0.75 (0.20-2.66)	0.68 (SE: 0.64)
TLR (main effect)	0.17 (0.06-0.52)	0.19 (SE: 0.53)
Adjustment Covariates (Ridge-unpenalized)		
Age	1.00 (0.96-1.03)	0.99 (SE: 0.02)
Sex	1.22 (0.62-2.45)	1.24 (SE: 0.35)
Rx (Experimental)	0.75 (0.20-2.66)	1.49 (SE: 0.55)
CRP	0.43 (0.18-0.96)	0.41 (SE: 0.43)
TMB/BRAF	0.50 (0.24-0.99)	0.49 (SE: 0.36)

Note: Ridge applies L2 shrinkage to the TLR main effect and the TLR x Rx interaction only; Age, sex, Rx, CRP, and TMB/B

Interpretation: The Firth HR is the best unbiased point estimate of the TLR x Rx interaction under penalized maximum likelihood. The ridge HR applies L2 shrinkage targeted at the TLR terms; if the ridge estimate is substantially closer to HR = 1.0 than the Firth estimate, the responder-stratified interaction is sensitive to regularization, consistent with a small-sample, partly tautological signal. Findings here are exploratory and should not be used for pre-treatment patient selection, since TLR is not measurable at baseline. They may motivate landmark or formal causal-mediation analyses in future work.

8 Landmark Analysis: TLR at Week 9

The responder analysis above treats TLR status as if it were known at baseline. It is not: TLR is measured at the first on-treatment CT scan, so a patient can only be classified TLR-positive or TLR-negative *after* surviving (and remaining progression-free) long enough to reach that scan. Conditioning survival on a post-baseline response therefore induces **guarantee-time (immortal-time) bias** — TLR-classified patients are, by construction, a more favourable risk set than the full randomized population.

A **landmark analysis** removes this bias by (i) fixing a landmark time, (ii) restricting to patients still at risk at the landmark, and (iii) measuring survival **from the landmark onward**. Here the landmark is set to **week 9**, the time of the second CT (the first on-treatment response assessment): TLR compares this scan to the baseline CT, so a patient’s TLR status does not exist until week 9. Placing the landmark at the measurement time ensures every patient in the cohort has actually reached the scan and has a defined TLR. Every retained patient is guaranteed to have reached the assessment, so the TLR x Rx contrast is no longer inflated by the survival required to be classifiable. This section **complements** the responder analysis above (which is retained for comparison); it does not replace it.

If the true response-assessment week differs from 9, only the single `LANDMARK_WK` value below needs changing. Because the small trial loses patients with events or censoring before the landmark, the cohort sizes and event counts are reported explicitly, and all model fits are wrapped so a too-small cohort degrades gracefully rather than aborting the render.

8.1 Landmark Cohort and Attrition

Table 17: Week-9 landmark cohort: attrition and remaining events

Endpoint	N randomized (complete cases)	N at risk at week 9	N excluded (event/censor < week 9)	Events from week 9
Overall survival	65	65	0	56
Progression-free survival	65	56	9	43

Note: Patients with death, progression, or censoring before week 9 are excluded from the corresponding endpoint. Survival is measured from week 9 onward.

8.2 Descriptive Context (Landmark Cohort)

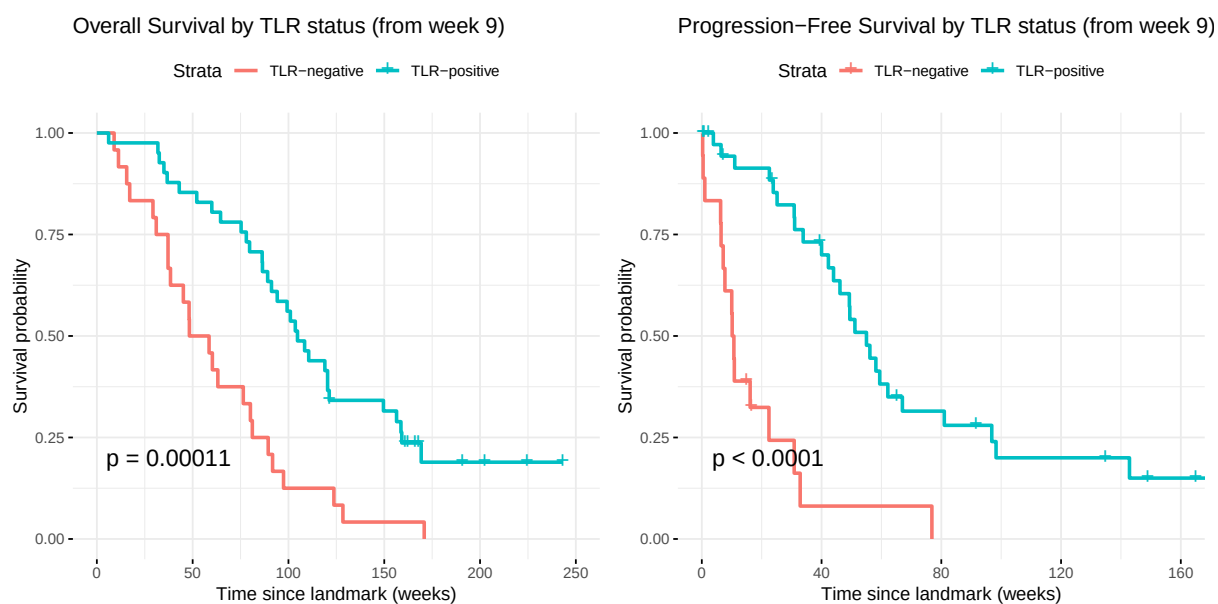
8.2.1 TLR Prevalence by Treatment Arm at Week 9

Table 18: TLR Prevalence by Treatment Arm, Week-9 Landmark Cohort (OS at-risk set)

Treatment Arm	N	TLR-positive	%
Control (FLOX)	29	22	75.9
Experimental (FLOX/nivo)	36	19	52.8

Note: Restricted to patients alive at week 9. TLR = Tumour Lesion Reduction $\geq 10\%$ at first CT scan.

8.2.2 Landmark Kaplan-Meier Curves Stratified by TLR



Note: Follow-up time is measured from the week-9 landmark. All patients shown were alive (OS) or alive and progression-free (PFS) at the landmark, so the curves are free of the guarantee-time bias affecting the responder-analysis curves above.

8.3 Standard Cox vs Firth (Landmark)

Table 19: Landmark (Week 9) TLR Model - Overall Survival: Standard Cox vs Firth

Term	Standard Cox HR (95% CI)	Firth HR (95% CI)
TLR Terms		
TLR x Rx	2.43 (0.74-7.95)	2.47 (0.75-7.75)
TLR (main effect)	0.21 (0.08-0.54)	0.21 (0.08-0.55)
Adjustment Covariates		
Rx (Experimental)	0.60 (0.23-1.55)	0.59 (0.24-1.55)
CRP	0.52 (0.25-1.06)	0.53 (0.25-1.05)
TMB/BRAF	0.86 (0.48-1.52)	0.86 (0.48-1.52)
Age	1.01 (0.98-1.04)	1.01 (0.98-1.04)
Sex	1.48 (0.81-2.68)	1.47 (0.82-2.67)

Table 20: Landmark (Week 9) TLR Model - Progression-Free Survival: Standard Cox vs Firth

Term	Standard Cox HR (95% CI)	Firth HR (95% CI)
TLR Terms		
TLR x Rx	0.86 (0.20-3.59)	0.88 (0.21-3.42)
TLR (main effect)	0.16 (0.05-0.53)	0.17 (0.05-0.55)
Adjustment Covariates		
Rx (Experimental)	1.08 (0.31-3.83)	1.05 (0.32-3.80)
CRP	0.52 (0.22-1.24)	0.53 (0.22-1.23)
TMB/BRAF	0.35 (0.16-0.77)	0.36 (0.16-0.77)
Age	1.00 (0.96-1.04)	1.00 (0.96-1.04)
Sex	1.06 (0.50-2.27)	1.06 (0.50-2.27)

8.4 Firth-Corrected Cox Model (Landmark)

8.4.1 Overall Survival

Table 21: Landmark TLR Model: Overall Survival - Full Coefficient Table

Term	HR (95% CI)	PLRT p-value
Treatment x Biomarker Interaction		
RxExperimental arm:tlr_num	2.47 (0.75-7.75)	0.135
Main Effects and Covariates		
Age	1.01 (0.98-1.04)	0.626
sex1	1.47 (0.82-2.67)	0.201
RxExperimental arm	0.59 (0.24-1.55)	0.270
crp_num	0.53 (0.25-1.05)	0.071
tmb_braf_num	0.86 (0.48-1.52)	0.609
tlr_num	0.21 (0.08-0.55)	0.002

8.4.2 Progression-Free Survival

Table 22: Landmark TLR Model: Progression-Free Survival - Full Coefficient Table

Term	HR (95% CI)	PLRT p-value
Treatment x Biomarker Interaction		
RxExperimental arm:tlr_num	0.88 (0.21-3.42)	0.851
Main Effects and Covariates		
Age	1.00 (0.96-1.04)	0.972
sex1	1.06 (0.50-2.27)	0.877
RxExperimental arm	1.05 (0.32-3.80)	0.934
crp_num	0.53 (0.22-1.23)	0.142
tmb_braf_num	0.36 (0.16-0.77)	0.008
tlr_num	0.17 (0.05-0.55)	0.005

8.5 Ridge Sensitivity Analysis (Landmark, TLR Terms Penalized)

Ridge regression is applied to the landmark cohorts with the same targeted penalty as the responder analysis: only the TLR main effect and the TLR x Rx interaction are L2-shrunk, while Age, sex, Rx, CRP, and TMB/BRAF remain unpenalized.

Table 23: Landmark TLR Model - Overall Survival: Firth vs Ridge, all coefficients

Term	Firth (95% CI)	Ridge HR (SE)
TLR Terms (Ridge-penalized)		
TLR x Rx	2.47 (0.75-7.75)	2.12 (SE: 0.58)
TLR (main effect)	0.21 (0.08-0.55)	0.24 (SE: 0.47)
Adjustment Covariates (Ridge-unpenalized)		
Age	1.01 (0.98-1.04)	1.01 (SE: 0.02)
Sex	1.47 (0.82-2.67)	1.49 (SE: 0.30)
Rx (Experimental)	2.47 (0.75-7.75)	0.66 (SE: 0.48)
CRP	0.53 (0.25-1.05)	0.52 (SE: 0.36)
TMB/BRAF	0.86 (0.48-1.52)	0.86 (SE: 0.29)

Note: Week-9 landmark cohort. Ridge applies L2 shrinkage to the TLR main effect and the TLR x Rx interaction only; Age, sex, Rx, CRP, and TMB/BRAF remain unpenalized.

Table 24: Landmark TLR Model - Progression-Free Survival: Firth vs Ridge, all coefficients

Term	Firth (95% CI)	Ridge HR (SE)
TLR Terms (Ridge-penalized)		
TLR x Rx	0.88 (0.21-3.42)	0.78 (SE: 0.68)
TLR (main effect)	0.17 (0.05-0.55)	0.18 (SE: 0.57)
Adjustment Covariates (Ridge-unpenalized)		
Age	1.00 (0.96-1.04)	1.00 (SE: 0.02)
Sex	1.06 (0.50-2.27)	1.08 (SE: 0.38)
Rx (Experimental)	0.88 (0.21-3.42)	1.17 (SE: 0.62)
CRP	0.53 (0.22-1.23)	0.52 (SE: 0.44)
TMB/BRAF	0.36 (0.16-0.77)	0.35 (SE: 0.40)

Note: Week-9 landmark cohort. Ridge applies L2 shrinkage to the TLR main effect and the TLR x Rx interaction only; Age, sex, Rx, CRP, and TMB/BRAF remain unpenalized.

Interpretation: The two endpoints behave very differently under the landmark, and the contrast is itself informative.

Overall survival is clean: 0 patient(s) are excluded (no deaths occur before week 9), so the landmark OS cohort is the full randomized population and the landmark Firth TLR x Rx HR (2.47 (0.75-7.75)) is **identical** to the responder estimate (2.47 (0.75-7.75)) — a Cox model is invariant to a common shift of the time origin when no one leaves the risk set. The OS responder signal is therefore not an early-death guarantee-time artefact.

Progression-free survival exposes a deeper problem. 9 patients are excluded, of whom 5 progressed, 5 of them **TLR-negative**. This is not a coincidence: progression and TLR are read from the *same* first on-treatment scan, so a patient whose tumour grows at that scan is simultaneously classified as a progression and as TLR-negative. The week-9 landmark removes exactly these TLR-negative scan-time progressors, shifting the Firth TLR x Rx HR from 0.75 (0.20-2.66) (responder) to 0.88 (0.21-3.42) (landmark). The responder PFS association is thus **partly tautological** — TLR-negativity and early progression are the same measurement — which the landmark makes explicit by excluding the coupled events.

Neither analysis resolves the most fundamental issue: TLR lies on the causal path of treatment (Rx $\rightarrow$ TLR $\rightarrow$ outcome), so the TLR x Rx contrast is not a baseline patient-selection estimate under any time origin; disentangling it would require formal causal mediation, not a landmark. All estimates remain exploratory and imprecise (PLRT p 0.135 OS, 0.851 PFS) and must not guide treatment selection. The landmark is fixed at week 9 (the second, first-on-treatment CT); if the assessment week differs, change `LANDMARK_WK`.

9 Discussion

9.1 Primary interaction results

Neither biomarker-treatment interaction reached conventional significance in the unified model. For **overall survival**, $\text{CRP} \times \text{Rx}$ yielded HR 0.78 (95% CI 0.21–3.61, PLRT $p = 0.728$) and $\text{TMB/BRAF} \times \text{Rx}$ HR 0.93 (0.30–2.85, $p = 0.893$) — point estimates close to null with very wide confidence intervals. For **progression-free survival**, the $\text{CRP} \times \text{Rx}$ interaction showed a more directionally suggestive effect (HR 0.33, 0.07–2.15, $p = 0.224$): CRP-positive patients in the experimental arm had roughly one-third the hazard of progression relative to CRP-positive controls, though the CI spans more than an order of magnitude. These results are consistent with the trial being underpowered to detect treatment-effect heterogeneity ($n = 65$, 48 progressions, 56 deaths): even a moderate subgroup effect (HR ~ 0.5) would require far larger samples for reliable estimation.

The PFS pattern is worth noting as a hypothesis-generating finding. The CRP direction (HR 0.33) is consistent with the DAG association test showing $\text{CRP} \rightarrow \text{PFS}$ (HR 0.40, $p = 0.004$) and the biological hypothesis that low CRP (reflecting lower systemic inflammation) identifies patients more likely to respond to immunotherapy. However, the OS CRP interaction estimate (HR 0.78) is considerably closer to null, suggesting that any PFS benefit does not clearly translate to an OS benefit in this dataset.

9.2 Proportional hazards and TMB/BRAF

The PH assumption was well supported in overall survival (Schoenfeld global $p = 0.40$). In progression-free survival, however, the global test was significant ($p = 0.007$), driven mainly by TMB/BRAF — both the main effect ($p = 0.021$) and the $\text{TMB/BRAF} \times \text{Rx}$ interaction ($p = 0.045$) violated the PH assumption. This indicates that the hazard ratio for TMB/BRAF (and its modification by treatment) is not constant over time in PFS, which may reflect different kinetics of early vs late events in TMB/BRAF-positive tumours. The PFS $\text{TMB/BRAF} \times \text{Rx}$ interaction estimate (HR 0.60) should therefore be interpreted with caution; a time-varying Cox model or landmark analysis would be more appropriate for that specific question.

9.3 Ridge regression sensitivity

For OS, Ridge shrinks the $\text{CRP} \times \text{Rx}$ estimate substantially toward null (Firth HR 0.78 $\rightarrow$ Ridge HR 0.39), suggesting the OS estimate is sensitive to regularization and should be treated as imprecise. The $\text{TMB/BRAF} \times \text{Rx}$ OS estimate shows minimal shrinkage (0.93 $\rightarrow$ 0.87), consistent with a near-null interaction. For PFS, the $\text{CRP} \times \text{Rx}$ estimate is essentially unchanged by Ridge (Firth 0.33 $\rightarrow$ Ridge 0.32), indicating that the directional PFS finding is robust to regularization — the data support it even when the model is penalized for large coefficients. $\text{TMB/BRAF} \times \text{Rx}$ PFS shows moderate shrinkage (0.60 $\rightarrow$ 0.45).

9.4 TLR responder analysis (exploratory)

The exploratory responder model — $\text{Surv} \sim \text{Age} + \text{sex} + \text{Rx} + \text{CRP} + \text{TMB/BRAF} + \text{TLR} + \text{TLR}:\text{Rx}$ — estimates the differential treatment effect between TLR-positive and TLR-negative subgroups while adjusting for the pre-treatment biomarkers as prognostic main effects. The Firth-estimated $\text{TLR} \times \text{Rx}$ HR was 2.47 (0.75–7.75) for overall survival and 0.75 (0.20–2.66) for progression-free survival. The OS estimate sits **above 1.0**, which directionally implies the (TLR-positive vs TLR-negative) hazard contrast is *worse* on the experimental arm than on the control arm — the opposite of a “TLR-positive predicts immunotherapy benefit” pattern. PFS is closer to null. Ridge shrinkage of the TLR terms gives 2.12 (SE: 0.58) (OS) and 0.68 (SE: 0.64) (PFS): ridge moves the OS estimate modestly toward null but preserves the directional

pattern, and barely changes the PFS estimate. PLRT p-values (0.135 OS, 0.660 PFS) do not reach conventional significance, consistent with the wide profile-likelihood CIs in this small sample.

The directional pattern is consistent with the descriptive imbalance: TLR-positive prevalence is 75.9% in the control arm and 52.8% in the experimental arm. If TLR cleanly captured immunotherapy response one would expect higher TLR-positivity in the experimental arm, not lower. The observed prevalence pattern, together with the OS direction ($HR > 1$) of the $TLR \times Rx$ interaction, is more consistent with TLR acting as a **prognostic** marker (identifying chemotherapy-responsive disease, which is more prevalent in the control arm by chance or by timing) than as a predictive marker for immunotherapy benefit. Possible explanations for the imbalance include small-sample randomization imbalance, a timing artefact (TLR measured at a fixed CT that may occur during different treatment phases across arms), or genuine heterogeneity in the patient mix. Because TLR is itself influenced by treatment, the $TLR \times Rx$ interaction is a responder-stratified contrast — useful for hypothesis generation, but not a causal predictive-biomarker estimate.

9.5 TLR landmark analysis (exploratory)

A **week-9 landmark analysis** conditions on survival to the second (first on-treatment) CT — the scan at which TLR is defined — and measures survival from that point, to test whether the responder signal is a guarantee-time artefact. The two endpoints diverge instructively. For **OS**, 0 patients are excluded (no early deaths), so the landmark estimate is identical to the responder estimate (2.47 (0.75-7.75)): the OS signal is not driven by early-death immortal time. For **PFS**, 9 patients are excluded, with 5 of the 5 excluded progressors being TLR-negative, because progression and TLR are ascertained at the same scan; removing these coupled events moves the Firth $TLR \times Rx$ HR from 0.75 (0.20-2.66) to 0.88 (0.21-3.42). The PFS responder association is therefore partly tautological. Under any time origin TLR remains on the causal path of treatment, so a definitive predictive-biomarker estimate would require causal mediation rather than a landmark. Both analyses remain underpowered and exploratory, and neither supports baseline treatment selection on TLR.

10 Summary

Study: METIMMOX-1 biomarker subgroup analysis. N = 65 complete cases, 56 OS events, 48 PFS events.

Methods: Two Firth-corrected Cox analyses are presented. **Primary (DAG-informed):** $\text{Surv} \sim \text{Age} + \text{sex} + \text{Rx} + \text{CRP} + \text{TMB/BRAF} + \text{CRP}:\text{Rx} + \text{TMB/BRAF}:\text{Rx}$ — pre-treatment biomarkers as effect modifiers; TLR omitted because it is post-randomization. **Exploratory responder analysis:** $\text{Surv} \sim \text{Age} + \text{sex} + \text{Rx} + \text{CRP} + \text{TMB/BRAF} + \text{TLR} + \text{TLR}:\text{Rx}$ — adds TLR and its treatment interaction, with CRP and TMB/BRAF retained as prognostic main-effect adjustments. Profile likelihood CIs, PLRT for interaction testing. Ridge sensitivity analysis penalizes the CRP/TMB main effects in the primary model and the TLR terms (main effect + TLR:Rx) in the exploratory model.

Proportional hazards: No violations in OS (global p = 0.40). In PFS, the TMB/BRAF main effect (p = 0.021) and TMB/BRAF $\times$ Rx interaction (p = 0.045) violate the PH assumption; PFS TMB/BRAF estimates should be interpreted with caution.

Interaction estimates:

Interaction	OS HR (95% CI)	PLRT p	PFS HR (95% CI)	PLRT p
CRP $\times$ Rx	0.78 (0.21–3.61)	0.728	0.33 (0.07–2.15)	0.224
TMB/BRAF $\times$ Rx	0.93 (0.30–2.85)	0.893	0.60 (0.16–2.28)	0.447

No interaction is statistically significant. CRP $\times$ Rx PFS (HR 0.33) is the most directionally consistent finding and is robust to Ridge regularization (Ridge HR 0.32). TMB/BRAF OS shows near-null interaction (Ridge HR 0.87 Firth); TMB/BRAF PFS violated PH.

TLR responder analysis (exploratory): Firth TLR $\times$ Rx HR 2.47 (0.75-7.75) for OS and 0.75 (0.20-2.66) for PFS. Ridge-penalized TLR terms: 2.12 (SE: 0.58) (OS), 0.68 (SE: 0.64) (PFS). PLRT p-values (0.135 OS, 0.660 PFS) do not reach conventional significance in this small sample. TLR-positive prevalence: control 75.9%, experimental 52.8% — markedly higher TLR-positivity in the control arm. The OS direction (HR > 1) of TLR $\times$ Rx, combined with the prevalence pattern, is more consistent with TLR acting as a prognostic marker for chemo-responsive disease than as a predictive marker for immunotherapy benefit. Because TLR is post-randomization, this is a responder-stratified, hypothesis-generating contrast rather than a causal predictive-biomarker estimate.

TLR week-9 landmark analysis (exploratory): A landmark analysis conditioning on survival to the second (first on-treatment) CT in week 9 — the scan at which TLR is defined — was added. For OS, 0 patients are excluded (no early deaths), so the landmark HR 2.47 (0.75-7.75) is identical to the responder value: the OS signal is not an early-death immortal-time artefact. For PFS, 9 patients are excluded, 5 of the 5 excluded progressors being TLR-negative because progression and TLR are read from the same scan; this shifts the HR from 0.75 (0.20-2.66) (responder) to 0.88 (0.21-3.42) (landmark), exposing the PFS association as partly tautological (PLRT p 0.135 OS, 0.851 PFS). Both analyses are retained for comparison; both are underpowered and exploratory, and the residual confounding from TLR being treatment-influenced needs causal mediation, not a landmark. Neither supports baseline treatment selection on TLR.

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